

Workforce Management System — Documentation Notes

Based on the uploaded Operations Dashboard system documentation and sample operational dataset.

1. Project Purpose

The project is a **Workforce Management (WFM) Operations Dashboard** designed for contact center / BPO environments such as Teleperformance.

The system monitors:

- Agent productivity
- Schedule adherence
- Leave management
- Break compliance
- Coverage availability
- SLA compliance
- Operational efficiency
- Financial risk

The core business objective is to ensure the company continuously delivers at least **95% contractual capacity** to the client. Falling below this level creates:

- SLA breaches
- Revenue loss
- Staffing shortages
- Operational penalties
- Customer experience degradation

The system acts as a **central operational intelligence platform** connecting four departments:

1. Operations
2. Workforce Management (WFM)
3. Finance
4. Recruitment

The Operations module is the heart of the system and focuses on:

- Working hours tracking
 - Annual leave control
 - Break and shrinkage management
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2. Business Problem the System Solves

Large support centers manage hundreds or thousands of agents.

Without automation:

- Supervisors cannot track all agents manually.
- Shrinkage increases unnoticed.
- Leave approvals can reduce staffing below SLA requirements.
- Agents may appear online but remain idle.
- Overtime and staffing shortages become reactive instead of predictive.

This system solves those problems by:

- Monitoring agent activity continuously
 - Detecting abnormal behaviour
 - Calculating staffing coverage
 - Predicting SLA risk
 - Providing live operational alerts
 - Supporting decision-making using analytics and AI
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3. High-Level System Architecture

The system follows a departmental workflow:

Recruitment → WFM → Operations → Finance

Each department contributes to maintaining the client SLA target.

4. Department Responsibilities

4.1 Recruitment Department

Recruitment ensures:

- Enough agents are hired
- Correct skill/tunnel assignment
- Replacement staffing for attrition

Main Responsibilities

- Hiring
- Skill assignment
- New hire onboarding
- Attrition replacement

Key Metrics

- Active headcount
 - Attrition rate
 - Time-to-productive (TTP)
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4.2 WFM (Workforce Management)

WFM manages:

- Scheduling
- Forecasting
- Capacity planning
- Staffing requirements

Main Responsibilities

- Forecast customer volume
- Allocate shifts
- Calculate required staffing
- Maintain service level

Key Metrics

- Service level
 - Occupancy
 - Forecast accuracy
 - Required FTEs
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4.3 Operations Department

Operations manages:

- Real-time workforce activity
- Agent adherence
- Productivity
- Leave approvals
- Shrinkage

Main Responsibilities

- Verify attendance
- Monitor breaks
- Track productive hours
- Ensure coverage

Key Metrics

- OCC %
 - Shrinkage %
 - Adherence %
 - Productive hours
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4.4 Finance Department

Finance validates:

- Billable hours
- Contract utilization
- Penalty exposure

Main Responsibilities

- Revenue compliance

- Billing validation
- Financial forecasting

Key Metrics

- Billable compliance
 - Penalty risk
 - Utilization
-

5. Complete Architecture Flow

Client Contract (95% SLA)



Recruitment → WFM → Operations → Finance

Flow Explanation

Department	What It Ensures
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Recruitment	Right agents
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WFM	Right schedule
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Operations	Right behaviour
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Finance	Right billing
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6. Core Functional Modules

The project contains 3 major operational modules.

Module 1 — Working Hours Management

Purpose:

Track whether agents worked their contracted hours properly.

The system compares:

- Scheduled hours
- Actual login duration

to determine:

- Adherence
 - Productivity
 - Overtime need
 - Compliance risk
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Important Concepts

Shift Span

Measures total time between:

- first activity
- last activity

Formula:

$$\text{Shift Hours} = \text{Last Activity Timestamp} - \text{First Login Timestamp}$$

Used to detect:

- late login
 - early logout
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Productive Hours

Actual working time after removing breaks.

$$\text{Productive Hours} = \text{Total Logged Hours} - (\text{Break Hours} + \text{Lunch Hours})$$

Example:

- 8h logged

- 50 min break/lunch

Productive = 7h 10m approx.

Adherence

Measures whether the agent followed the planned schedule.

High adherence means:

- logged in on time
- available when expected
- following assigned states

Low adherence causes:

- SLA risk
 - understaffing
 - penalty exposure
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Overtime Trigger

If the organization delivers below 95% projected hours:

- overtime may be triggered
 - backfill staffing may be required
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Module 2 — Annual Leave Management

Purpose:

Approve employee leave without reducing operational coverage below SLA requirements.

Coverage Rule

The company must maintain at least:

- 95% staffing coverage

Example:

- Team size = 20
- Max leave allowed simultaneously = 1 agent

If more agents take leave:

- coverage drops below SLA
 - penalty risk increases
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Leave Logic

The leave system checks:

1. Leave balance
2. Team coverage
3. Blackout dates
4. Notice period

before approval.

Important Leave Formula

$$\text{Coverage \%} = \frac{\text{Active Agents}}{\text{Required Agents}} \times 100$$

Leave Approval Workflow

The system:

- denies leave if coverage falls below 95%
- prevents approvals during blackout dates
- validates sufficient notice period

Pseudo-logic exists in the uploaded documentation.

Module 3 — Break & Shrinkage Management

Purpose:

Control paid non-productive time.

What is Shrinkage?

Shrinkage = paid time where the agent is unavailable for customer handling.

Examples:

- breaks
- lunch
- training
- meetings

Shrinkage directly reduces:

- productivity
 - billable hours
 - staffing efficiency
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Shrinkage Formula

$$\text{Shrinkage \%} = \frac{\text{Break Minutes} + \text{Lunch Minutes}}{\text{Total Minutes}} \times 100$$

Occupancy Formula

Measures how busy agents are while available.

$$\text{Occupancy \%} = \frac{\text{OCC}}{\text{OCC} + \text{CFT}} \times 100$$

Staffing Formula

WFM uses shrinkage to calculate actual required staffing.

$$\text{Required Seats} = \frac{\text{Base FTEs}}{1 - \text{Shrinkage \%}}$$

Example:

- Need 20 productive agents
- Shrinkage = 20%

Actual required roster:

- 25 agents
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7. Status Groups Explained

The CSV dataset classifies activities into groups.

OCC (Occupied)

Most valuable state.

Agent is actively handling customer work.

Examples:

- chat
- after_contact_work (ACW)

This is:

- directly billable
 - revenue-generating time
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CFT (Client Facing Time)

Agent is available and ready.

Examples:

- available
- email
- small_talking

The agent is productive but not actively chatting.

Shrinkage

Non-productive paid time.

Examples:

- break
- lunch

Must remain controlled.

UTI / UT (Utilisation)

Training or meetings.

Small amounts are healthy because:

- agents improve skills
 - onboarding happens
 - coaching occurs
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8. Important Engineering Concepts

SLA (Service Level Agreement)

A client contract defining required operational performance.

Example:

- Maintain 95% staffing capacity

If breached:

- penalties apply
 - contract risk increases
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FTE (Full-Time Equivalent)

Represents one full employee workload.

Example:

- 1 FTE = 8-hour agent shift
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Erlang-C

A mathematical staffing model used in contact centers.

Purpose:

Calculate how many agents are needed based on:

- forecasted contacts
 - handling time
 - occupancy target
 - shrinkage
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AHT (Average Handle Time)

Average time to complete one customer interaction.

Formula:

$$\text{AHT} = \frac{\text{Talk Time} + \text{ACW}}{\text{Number of Contacts}}$$

ACW (After Contact Work)

Time spent after customer interaction:

- notes
- ticket updates
- documentation

Still considered productive.

Coverage

Percentage of active agents compared to required agents.

High coverage:

- good SLA protection

Low coverage:

- staffing risk
 - longer wait times
 - penalty exposure
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Occupancy

Measures how busy agents are.

Low occupancy:

- overstaffing

Very high occupancy:

- burnout risk
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9. AI & Analytics Features

The documentation proposes multiple AI capabilities.

Statistical AI Features

Easy to implement:

- anomaly detection
- trend analysis
- rolling averages
- threshold alerts

Example:

- flag agent OCC below 50%
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Machine Learning Features

Medium difficulty:

- attrition prediction
- shrinkage forecasting
- staffing prediction

Requires:

- historical datasets
 - training pipeline
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Real-Time Features

Examples:

- live coverage monitor
- SLA breach prediction
- stuck-status detection

Uses:

- WebSockets
 - streaming aggregation
 - real-time processing
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LLM Features

Examples:

- natural-language daily summaries
 - conversational analytics
 - AI coaching recommendations
 - leave reasoning assistant
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10. Dataset Observations

The uploaded sample dataset identified:

- 20 agents
- 195.5 tracked hours
- 48% OCC
- 25.3% shrinkage

Important findings:

- abnormal idle agents
- potential attrition indicators
- inconsistent shrinkage
- missing legal breaks
- different shift patterns

These observations show why automated monitoring is required.

11. Missing Data Required for Production

The documentation states the CSV alone is insufficient for full production deployment.

Missing critical data:

- scheduled shifts
 - contracted hours
 - leave balances
 - contact counts
 - queue targets
 - HR mapping
 - forecast data
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12. Recommended Technical Architecture

Based on the documentation and your preferred stack:

Frontend

- JavaFX dashboard
- Charts and live monitoring
- Supervisor console

Backend

- FastAPI
- REST + WebSocket APIs
- AI/ML services

Data Layer

- CSV initially
- Later PostgreSQL/data warehouse

Communication

- gRPC for internal service communication

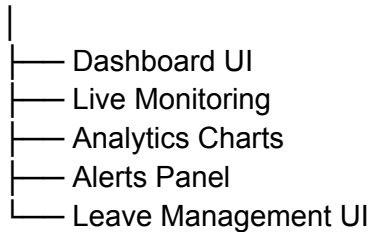
- WebSockets for real-time monitoring

AI/ML Stack

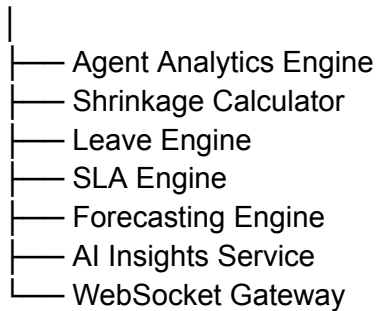
- Pandas
 - scikit-learn
 - Prophet
 - XGBoost
 - LLM APIs
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13. Recommended Project Structure

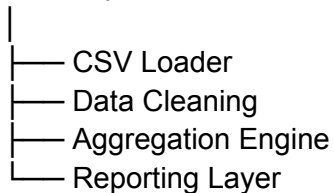
Frontend (JavaFX)



Backend (FastAPI)



Data Layer



14. Recommended Development Phases

Phase 1

Basic analytics and rules.

Phase 2

LLM-generated insights.

Phase 3

Predictive ML models.

Phase 4

Optimization and intelligent scheduling.

15. Final Understanding

This project is not just a dashboard.

It is a:

- workforce intelligence platform
- operational monitoring system
- staffing optimization engine
- SLA compliance system
- financial risk management platform
- AI-assisted operational analytics solution

Its ultimate purpose is:

Deliver the required workforce capacity at minimum operational cost while maintaining contractual SLA compliance.